



1. High-Pressure Test Pumps & Fluid Systems





Hydrostatic & Pneumatic Pressure Testing Systems

Advanced High-Pressure Engineering & Turnkey Testing Infrastructure

OWU designs, fabricates, and supplies advanced hydrostatic and pneumatic pressure testing systems engineered for elite performance, total structural integrity, and uncompromising safety. We deliver high-precision testing infrastructure for pipelines, valves, pressure vessels, heat exchangers, high-pressure hoses, tubing, cylinders, and critical oilfield process equipment.

By pairing heavy-duty mechanical fabrication with state-of-the-art pressure instrumentation, automated data acquisition (DAQ), and programmable logic control (PLC) architectures, OWU provides turnkey, compliance-ready systems. All solutions are custom-engineered to align with strict customer specifications and international testing standards, including **API, ASME, and ISO**.

Core Products & Systems Integration

1. High-Pressure Hydrotest Bays & Blast Enclosures

OWU designs and constructs permanent Hydrotest Pressure Bays engineered to provide an elite, controlled environment for high-risk hydrostatic and pneumatic testing operations.

- **Structural Safety:** Heavy-duty reinforced steel construction, blast-resistant multi-layered test enclosures, and automated safety interlock systems that prevent bay entry while pressurized.
- **System Automation:** High-pressure pumping systems integrated with automated test sequences, pneumatic isolation valves, and remote operation capabilities to remove personnel from the live testing zone.
- **Advanced Control:** Features multi-sensor arrays, proportional control valves, and integrated digital pressure monitoring for smooth, stepped pressure ramping.
- **Applications:** Production valve testing, pressure vessel qualification, pipeline component verification, and high-pressure oilfield equipment certification.

2. Containerized Mobile Pressure Test Systems



Our Containerized Pressure Test Systems package a fully self-contained, high-pressure testing facility inside a rugged, standard ISO container—engineered specifically for rapid deployment to remote oilfields and harsh environments.

- **Mobility & Durability:** Fully structural ISO container modifications featuring integrated climate-controlled operator control rooms, heavy-duty industrial flooring, robust internal lighting, and auxiliary power distribution.
- **Operational Readiness:** Rapid deployment and commissioning capabilities, making them highly suited for pipeline hydrotesting, offshore platforms, refinery shutdowns, and remote EPC construction projects.
- **Safety Isolation:** Built-in blast wall partitions that visually and physically isolate the operator workstation from the high-pressure pumping and manifold zone.

3. Portable & Field Service Test Units

Engineered for field maintenance, rig-site testing, and rapid-response operations, our Portable Pressure Test Systems deliver robust, high-pressure generation in a highly maneuverable format.

- **Power Versatility:** Available in compact, heavy-duty wheeled skids or crash-framed configurations driven by electric motors, pneumatic air-driven liquid pumps, or diesel engines.

- **Precision Control:** Outfitted with rugged digital pressure gauges, high-accuracy manual needle valves, and multi-port manifolds for testing multiple lines simultaneously.
- **Applications:** On-site equipment recertification, pipeline maintenance, wellhead valve verification, and field commissioning activities.

Digital Instrumentation, Telemetry & Data Logging

Accurate monitoring and unalterable test documentation are critical for strict regulatory compliance and quality assurance. OWU integrates premium digital hardware and legacy chart tracking to cover all verification preferences.

Digital Pressure Recording Systems & SCADA

Our digital suites replace traditional manual logging with automated, real-time data acquisition systems (DAQ) that deliver high-accuracy pressure measurement and traceable test certification.

- **Automation & HMIs:** PLC-based automation systems paired with touchscreen Human-Machine Interfaces (HMI) for real-time graphical pressure-trend displays and automated leak detection analysis.
- **Smart Features:** Multi-channel data acquisition, secure local data storage, network connectivity for remote engineering review, and automated PDF test report generation.
- **Sensors & Telemetry:** Industrial-grade, high-accuracy pressure transmitters, digital pressure calibrators, temperature monitoring loops, and wireless telemetry options for long-distance monitoring.

Norma Chart Recorders

For operations requiring permanent mechanical records and traditional traceable test documentation, we supply and seamlessly integrate industry-proven **Norma Chart Recorders**.

- **Traceability:** High-precision, continuous physical chart recording for hydrostatic and pneumatic testing.
- **Field Reliability:** Fully mechanical or battery-backed options designed to withstand harsh field environments, ensuring compliance-ready documentation for regulatory inspections.

Technical Specifications & System Components

Our testing systems utilize a rigorous selection of premium high-pressure components to guarantee continuous uptime and safe operation under extreme cyclical loading:

System Component	Specifications & Engineering Standard

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High-Pressure Pumps	Air-driven liquid pumps (e.g., Haskel/Maximator or equivalent) and electric-driven triplex plunger pumps capable of generating up to 2,000+ bar (30,000+ PSI) .
Piping & Manifolds	Autoclave-style medium and high-pressure cone-and-thread fittings, heavy-wall duplex or 316/316L stainless steel tubing, and high-pressure rated needle/ball valves.
Automation & Controls	Industrial Siemens/Allen-Bradley PLCs, high-resolution touch HMIs, and custom SCADA software scripts featuring automated pressure hold-time alerts and safety shutdown protocols.
Instrumentation Accuracy	Pressure transmitters and digital test gauges featuring up to ±0.05% Full Scale (FS) accuracy, with valid calibration certificates traceable to international bodies.
Pneumatic Testing Modules	Integrated high-pressure gas boosters for specialized gas/pneumatic testing, complete with automated venting systems and nitrogen bottle manifold racks.

Comprehensive Engineering & Fabrication Capabilities

OWU provides single-source accountability. We manage the entire lifecycle of your pressure testing infrastructure through a specialized in-house workflow:

- **System Design & Engineering:** Custom 3D CAD modeling, finite element analysis (FEA) for blast walls, hydraulic circuit schematics, and P&ID development.
- **Mechanical & Structural Fabrication:** Heavy structural steel welding, enclosure fabrication, and precision pipefitting executed by certified technicians.
- **Electrical Control Panels:** In-house assembly of certified electrical control panels, PLC programming, instrumentation calibration, and functional loop testing.
- **Testing & Quality Assurance:** Rigorous Factory Acceptance Testing (FAT) at our facility followed by Site Acceptance Testing (SAT), leveling, installation, and field commissioning.
- **Operator Training:** On-site technical training covering system operations, high-pressure safety protocols, software data logging, and routine maintenance calibration.

Industries We Serve

Our high-pressure testing solutions support critical infrastructure, manufacturing, and service companies worldwide:

- **Oil & Gas / Oilfield Service Providers:** High-pressure test bays for blowout preventers (BOPs), Christmas trees, and downhole tools.
- **Refineries & Petrochemical Facilities:** Shutdown testing equipment, heat exchanger bundle verification, and valve recertification units.
- **EPC Contractors & Construction:** Large-scale pipeline hydrotesting manifolds and containerized mobile field stations.
- **Marine, Offshore & Heavy Utilities:** Power generation pressure loop certification and ruggedized offshore-ready test skids.

The OWU Advantage

- **Engineering Excellence:** Custom-engineered systems built around your specific dimensional constraints, pressure requirements, and target workflows.
- **Safety-First Culture:** Multi-tiered safety features—including blast walls, safety interlocks, and automated emergency vent valves—are integrated into the fundamental design of every system.
- **Turnkey Execution:** From initial concept and structural engineering to software integration, onsite commissioning, and technician training.
- **Dedicated After-Sales Support:** Rapid deployment for system calibration, preventative maintenance, critical spare parts supply, and software upgrades.

Engineering Consultation & Quotes

Whether your facility requires a fixed high-pressure hydrotest bay, a mobile containerized testing station, or a specialized digital data logging upgrade, our engineering team can design a system to match your operational parameters.

Contact OWU today to consult with a technical specialist, review your testing criteria, and receive a detailed engineering proposal.